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CYSE 476
4/16/26
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Project Report 3

Task 1: Using Big Number

```
// sam juarez 4/16/26 cyse 476 lab 3
// task1.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2dec(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *a = BN_new();
    BIGNUM *b = BN_new();
    BIGNUM *n = BN_new();
    BIGNUM *res = BN_new();

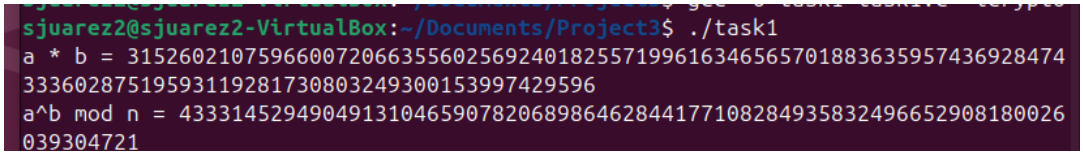
    BN_generate_prime_ex(a, 256, 1, NULL, NULL, NULL);
    BN_dec2bn(&b, "27348946379683850184859276946719436926 8");
    BN_rand(n, 256, 0, 0);

    BN_mul(res, a, b, ctx);
    printBN("a * b =", res);

    BN_mod_exp(res, a, b, n, ctx);
    printBN("a^b mod n =", res);

    BN_free(a); BN_free(b);
    BN_free(n); BN_free(res); BN_CTX_free(ctx);
    return 0;
}
```

Task1.c output:



Task 2: Deriving the Private Key

```
// sam juarez 4/16/26 cyse 476 lab 3
// task2.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2hex(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *p = BN_new();
    BIGNUM *q = BN_new();
    BIGNUM *e = BN_new();
    BIGNUM *p1 = BN_new();
    BIGNUM *q1 = BN_new();
    BIGNUM *phi = BN_new();
    BIGNUM *d = BN_new();
    BIGNUM *one = BN_new();

    BN_hex2bn(&p, "F7E75FDC469067FFDC4E847C51F452DF");
    BN_hex2bn(&q, "E85CED54AF57E53E092113E62F436F4F");
    BN_hex2bn(&e, "0D88C3");

    BN_dec2bn(&one, "1");
    BN_sub(p1, p, one);
    BN_sub(q1, q, one);
    BN_mul(phi, p1, q1, ctx);

    BN_mod_inverse(d, e, phi, ctx);
    printBN("private key d =", d);

    BN_free(p); BN_free(q); BN_free(e); BN_free(p1); BN_free(q1);
    BN_free(phi); BN_free(d); BN_free(one); BN_CTX_free(ctx);
    return 0;
}
```

Task2.c output

```
sjuarez2@sjuarez2-VirtualBox:~/Documents/Project3$ ./task2
private key d = 3587A24598E5F2A21DB007D89D18CC50ABA5075BA19A33890FE7C28A9B496AEB
```

Task 3: Encrypting a Message

```
// sam juarez 4/16/26 cyse 476 lab 3
// task3.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2hex(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *n = BN_new();
    BIGNUM *e = BN_new();
    BIGNUM *M = BN_new();
    BIGNUM *C = BN_new();

    BN_hex2bn(&n,
"DCBFFE3E51F62E09CE7032E2677A78946A849DC4CDDE3A4D0CB81629242FB1A5");
    BN_hex2bn(&e, "010001");
    BN_hex2bn(&M, "4120746f702073656372657421");

    BN_mod_exp(C, M, e, n, ctx);
    printBN("encrypted message C =", C);

    BN_free(n); BN_free(e); BN_free(M); BN_free(C); BN_CTX_free(ctx);
    return 0;
}
```

Task3.c output

```
sjuarez2@sjuarez2-VirtualBox:~/Documents/Project3$ ./task3
encrypted message C = 6FB078DA550B2650832661E14F4F8D2CFAEF475A0DF3A75CACDC5DE5CF
C5FADC
```

Task 4: Decrypting a Message

```
// sam juarez 4/16/26 cyse 476 lab 3
// task4.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2hex(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

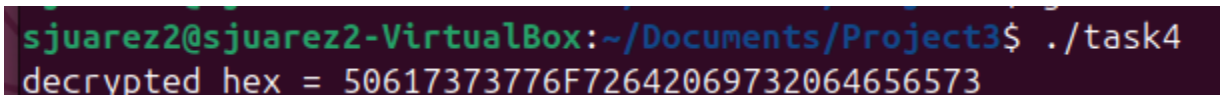
int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *n = BN_new();
    BIGNUM *d = BN_new();
    BIGNUM *C = BN_new();
    BIGNUM *M = BN_new();

    BN_hex2bn(&n,
"DCBFFE3E51F62E09CE7032E2677A78946A849DC4CDDE3A4D0CB81629242FB1A5");
    BN_hex2bn(&d,
"74D806F9F3A62BAE331FFE3F0A68AFE35B3D2E4794148AACBC26AA381CD7D30D");
    BN_hex2bn(&C,
"8C0F971DF2F3672B28811407E2DABBE1DA0FEBBBDFC7DCB67396567EA1E2493F");

    BN_mod_exp(M, C, d, n, ctx);
    printBN("decrypted hex =", M);

    BN_free(n); BN_free(d); BN_free(C); BN_free(M); BN_CTX_free(ctx);
    return 0;
}
```

Task4.c output



Task 5: Signing a Message

```
// sam juarez 4/16/26 cyse 476 lab 3
// task5.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2hex(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *n = BN_new();
    BIGNUM *d = BN_new();
    BIGNUM *M1 = BN_new();
    BIGNUM *M2 = BN_new();
    BIGNUM *S1 = BN_new();
    BIGNUM *S2 = BN_new();
    //$2000 and $3000 hex respectively \/
    BN_hex2bn(&n,
"DCBFFE3E51F62E09CE7032E2677A78946A849DC4CDDE3A4D0CB81629242FB1A5");
    BN_hex2bn(&d,
"74D806F9F3A62BAE331FFE3F0A68AFE35B3D2E4794148AACBC26AA381CD7D30D");

    BN_hex2bn(&M1, "49206f776520796f752024323030302e");
    BN_hex2bn(&M2, "49206f776520796f752024333030302e");

    BN_mod_exp(S1, M1, d, n, ctx);
    BN_mod_exp(S2, M2, d, n, ctx);

    printBN("signature of $2000 = ", S1);
    printBN("signature of $3000 = ", S2);

    BN_free(n); BN_free(d); BN_free(M1); BN_free(M2);
    BN_free(S1); BN_free(S2); BN_CTX_free(ctx);
    return 0;
}
```

Task5.c output

```
sjuarez2@sjuarez2-VirtualBox:~/Documents/Project3$ ./task5
signature of $2000 =  55A4E7F17F04CCFE2766E1EB32ADDBA890BBE92A6FBE2D785ED6E73CCB
35E4CB
signature of $3000 =  BCC20FB7568E5D48E434C387C06A6025E90D29D848AF9C3EBAC0135D99
305822
```

Task 6: Verifying a Signature

```
// sam juarez 4/16/26 cyse 476 lab 3
// task6.c
#include <stdio.h>
#include <openssl/bn.h>

void printBN(char *msg, BIGNUM *a){
    char *number_str = BN_bn2hex(a);
    printf("%s %s\n", msg, number_str);
    OPENSSL_free(number_str);
}

int main(){
    BN_CTX *ctx = BN_CTX_new();
    BIGNUM *n = BN_new();
    BIGNUM *e = BN_new();
    BIGNUM *S = BN_new();
    BIGNUM *M_rec = BN_new();

    BN_hex2bn(&n,
"AE1CD4DC432798D933779FBD46C6E1247F0CF1233595113AA51B450F18116115");
    BN_hex2bn(&e, "010001");
    BN_hex2bn(&S,
"643D6F34902D9C7EC90CB0B2BCA36C47FA37165C0005CAB026C0542CBDB6802F");

    BN_mod_exp(M_rec, S, e, n, ctx);
    printBN("recovered message hex = ", M_rec);

    BN_hex2bn(&S,
"643D6F34902D9C7EC90CB0B2BCA36C47FA37165C0005CAB026C0542CBDB6803F");
    BN_mod_exp(M_rec, S, e, n, ctx);
    printBN("recovered from corrupted signature = ", M_rec);

    BN_free(n); BN_free(e); BN_free(S); BN_free(M_rec);
BN_CTX_free(ctx);
    return 0;
}
```

Task6.c output

```
sjuarez2@sjuarez2-VirtualBox:~/Documents/Project3$ ./task6
recovered message hex =  4C61756E63682061206D697373696C652E
recovered from corrupted signature =  91471927C80DF1E42C154FB4638CE8BC726D3D66C8
3A4EB6B7BE0203B41AC294
```